

OPERATION ANALYTICS & INVESTIGATING METRIC SPIKE

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ANALYZING OPERATIONAL DATA TO IDENTIFY
TRENDS AND RESOLVE SPIKES



PROJECT OVERVIEW

DESCRIPTION

This project focuses on analyzing operational data to identify trends, detect anomalies, and investigate sudden changes in performance metrics. The objective was to simulate the role of a data analyst supporting business teams by delivering insights related to system throughput, user engagement, and platform health.

OBJECTIVE

- Analyze **job processing data** to track daily and hourly performance trends.
- Investigate **metric spikes** across user engagement, device usage, and email activity.
- Assist cross-functional teams (operations, support, marketing) in identifying bottlenecks, performance anomalies, and optimization opportunities.

TECH STACK USED

MySQL Workbench



CASE STUDY 1: JOB DATA ANALYSIS

A. JOBS REVIEWED OVER TIME

```
select ds as dates, round((count(job_id)/sum(time_spent))*3600) as "jobs reviewed per hour"
from job_data
where ds between '2020-11-01' and '2020-11-30'
group by ds;
```

	dates	jobs reviewed per hour
▶	2020-11-30	180
	2020-11-29	180
	2020-11-28	218
	2020-11-27	35
	2020-11-26	64
	2020-11-25	80

JOBS REVIEWED: MAX ON
NOVEMBER 28, 2020 (218 JOBS)



CASE STUDY 1: JOB DATA ANALYSIS

B.1:7 DAY ROLLING AVERAGE OF THROUGHPUT

```
WITH daily AS (  
  SELECT ds,  
    COUNT(job_id) AS daily_throughput  
  FROM job_data  
  GROUP BY ds  
)  
SELECT ds,  
  AVG(daily_throughput) OVER (  
    ORDER BY ds  
    ROWS BETWEEN 6 PRECEDING AND CURRENT ROW  
  ) AS rolling_7d_avg  
FROM daily  
ORDER BY ds;
```

THROUGHPUT ANALYSIS :
THROUGHPUT REMAINED STABLE
WITH MINOR, NON-DIRECTIONAL
FLUCTUATIONS.
TEAM PERFORMANCE IS STEADY BUT
NOT IMPROVING, INDICATING
CONSISTENT EXECUTION WITHOUT
CAPACITY GAINS.

	ds	rolling_7d_avg
▶	2020-11-25	1.0000
	2020-11-26	1.0000
	2020-11-27	1.0000
	2020-11-28	1.2500
	2020-11-29	1.2000
	2020-11-30	1.3333



CASE STUDY 1: JOB DATA ANALYSIS

B.2 : AVERAGE DAILY METRIC THROUGHPUT.

```
• SELECT AVG(daily_throughput) AS avg_daily_throughput
  FROM (
    SELECT ds, COUNT(job_id) AS daily_throughput
    FROM job_data
    GROUP BY ds
  ) t;
```

	avg_daily_throughput
▶	1.3333

FINDINGS:
ON AVERAGE, THE TEAM COMPLETES ABOUT 1.33 JOBS PER DAY, REFLECTING A LOW BUT CONSISTENT THROUGHPUT LEVEL.



CASE STUDY 1: JOB DATA ANALYSIS

C. LANGUAGE SHARE ANALYSIS

```
select `language`, count(job_id) as no_of_jobs,  
count(job_id)*100 / sum(count(*))over () as percentage  
from job_data  
where ds between '2020-11-01' and '2020-11-30'  
group by `language`;
```

	language	no_ofjobs	percentage
▶	English	1	12.5000
	Arabic	1	12.5000
	Persian	3	37.5000
	Hindi	1	12.5000
	French	1	12.5000
	Italian	1	12.5000

FINDINGS:
PERSIAN LANGUAGE IS THE
HIGHEST WITH 37.5% TOTAL.



CASE STUDY 1: JOB DATA ANALYSIS

D. DUPLICATION ROWS DETECTION

- `select * from`
 `( select *, row_number() over(partition by job_id) as row_num`
 `from job_data) a`
 `where row_num > 1;`

	ds	job_id	actor_id	event	language	time_spent	org	row_num
▶	2020-11-28	23	1005	transfer	Persian	22	D	2
	2020-11-26	23	1004	skip	Persian	56	A	3



FINDINGS: ROW WITH JOB_ID 23 WAS TAKEN TWO TIMES

CASE STUDY 2: INVESTIGATING METRIC SPIKES

A. FOR WEEKLY USER ENGAGEMENT

	week	weekly_user_engagement
▶	17	85
	18	194
	19	208
	20	195
	21	208
	22	230
	23	224
	24	252
	25	135
	26	21
	27	8
	28	5
	29	4
	30	1
	31	2

```
select
extract(week from occurred_at) as 'week',
count(distinct user_id) as 'weekly_user_engagement'
from events
group by extract(week from occurred_at);
```

FINDINGS: USER ENGAGEMENT SAW AN INCREASE FROM WEEK 18 TO WEEK 25, AFTER WHICH IT BEGAN TO DECLINE



CASE STUDY 2: INVESTIGATING METRIC SPIKES

B. USER GROWTH ANALYSIS

```
SELECT
  DATE(created_at) AS date,
  COUNT(*) AS daily_new_users,
  SUM(COUNT(*)) OVER (ORDER BY DATE(created_at)) AS total_users
FROM users
GROUP BY DATE(created_at)
ORDER BY date;
```

	date	daily_new_users	total_users
▶	2013-01-01	13	13
	2013-01-02	11	24
	2013-01-03	14	38
	2013-01-04	11	49
	2013-01-05	3	52
	2013-01-06	4	56
	2013-01-07	13	69
	2013-01-08	13	82
	2013-01-09	11	93
	2013-01-10	12	105
	2013-01-11	11	116



FINDINGS : OVERALL GROWTH IS HEALTHY AND ACCELERATING, BUT ACQUISITION IS EVENT-DRIVEN RATHER THAN EVENLY DISTRIBUTED ACROSS DAYS.



CASE STUDY 2: INVESTIGATING METRIC SPIKES

C. WEEKLY ENGAGEMENT PER DEVICE

```
select week(occured_at) as weeks,  
device,  
count(distinct user_id) as user_engagement from `events`  
group by device,  
week (occured_at)  
order by week(occured_at);
```

DEVICE INSIGHTS: MACBOOK &
IPHONE USERS MOST ACTIVE

Weeks	device	User_engagement
17	acer aspire desktop	12
17	acer aspire notebook	23
17	amazon fire phone	4
17	asus chromebook	23
17	dell inspiron desktop	20
17	dell inspiron notebook	48
17	hp pavilion desktop	17
17	htc one	19
17	ipad air	29
17	ipad mini	19
17	iphone 4s	27
17	iphone 5	69
17	iphone 5s	47
17	kindle fire	6
17	lenovo thinkpad	94
17	mac mini	7
17	macbook air	61
17	macbook pro	154



CASE STUDY 2: INVESTIGATING METRIC SPIKES

D. EMAIL ENGAGEMENT ANALYSIS

```
select
  week(occured_at) as week,
  count(distinct ( case
    when `action`= 'sent_weekly_digest' then user_id
  end)) as weekly_digest,
  count(distinct ( case
    when `action` = 'sent_reengagement_email' then user_id
  end)) as reengagement_mail,
  count(distinct ( case
    when `action` = 'email_open' then user_id
  end )) as opened_email,
  count(distinct (case
    when action = 'email_clickthrough' then user_id
  end)) as email_clickthrough
from emails
group by week(occured_at)
order by week(occured_at);
```

FINDINGS:

- THE EMAIL OPENING RATE IS AROUND 34% AND EMAIL CLICKING RATE IS AROUND 15%.
- THE USERS ARE ENGAGING WITH THE EMAIL SERVICE WHICH IS GOOD FOR THE COMPANY TO EXPAND.

Week	weekly_digest	reengagement_mail	opened_email	email_clickthrough
17	908	73	310	166
18	2602	157	900	425
19	2665	173	961	476
20	2733	191	989	501
21	2822	164	996	436
22	2911	192	965	478
23	3003	197	1057	529
24	3105	226	1136	549
25	3207	196	1084	524
26	3302	219	1149	550
27	3399	213	1207	613
28	3499	213	1228	594
29	3592	213	1201	583
30	3706	231	1363	625
31	3793	222	1338	444





KEY INSIGHTS

- Rolling metrics are better to smooth out daily fluctuations
- Metric spikes often correlate with user behavior patterns
- Device preference insights help optimize app experience
- Email engagement shows opportunity for CRM expansion



LEARNINGS

- Mastered writing **efficient SQL** for daily/weekly metrics
- Understood importance of **metric context** and smoothing
- Gained experience in **root cause analysis** through spikes
- Strengthened **operational visibility** through **structured queries**



BUSINESS IMPACT

- Helps proactively detect system failures or performance drops
- Enables real-time tracking of operational KPIs
- Prevents downtime by early anomaly detection
- Optimizes decision-making with spike awareness



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THANK YOU

